# FCFS Appointment Simulation: Behavior Sensitivity Report

CUIMC Appointment Simulation

May 7, 2026

## 1 Purpose and Main Questions

This report studies the first appointment policy in the simulation: first-come, first-served (FCFS). The goal is to understand how patient behavior parameters affect clinic performance and whether one patient class is helped more than the other.

The report is organized by behavior, not by metric. For each behavior we ask:

1. What parameter changes?
2. What stays fixed?
3. What happens overall?
4. What happens to Class 1?
5. What happens to Class 2?
6. Is the effect mainly efficiency, redistribution, or both?

The baseline uses `configs/baseline.yaml`: 32 slots per day, a 14-day booking horizon, 365 measured days, and two equal-size arrival streams with $\lambda_1 = 50$ and $\lambda_2 = 50$ arrivals per day. Both classes have the same baseline behavior parameters.

## 2 Simple Metric Definitions

The report uses rates and averages, not total counts. This keeps scenarios comparable even when simulation length or arrival volume changes.

| Plain name | Code name | Meaning |
|---|---|---|
| Served rate, Class $i$ | `percent_serviced_i` | Share of Class $i$ arrivals that complete service. This is the main class-level access measure. |
| Overall served rate | `overall_percent_serviced` | Share of all arrivals that complete service. This is the main aggregate access measure. |
| Utilization | `average_utilization` | Average share of daily slots that become completed visits. This is aggregate only; there is no class-specific utilization. |
| Accepted wait | `mean_accepted_booking_delay` | Average booking delay among patients who accepted an appointment. This can be affected by selection because patients who balk are excluded. |
| Offered wait | `mean_offered_booking_delay` | Average booking delay among patients who received an offer, including patients who balked. This is the better patient-facing wait measure. |
| Class gap | `percent_serviced_1 - percent_serviced_2` | Positive means Class 1 has higher served rate. Negative means Class 2 has higher served rate. |

Table 1: Simple metric names used in the report.

## 3 Baseline Scenario

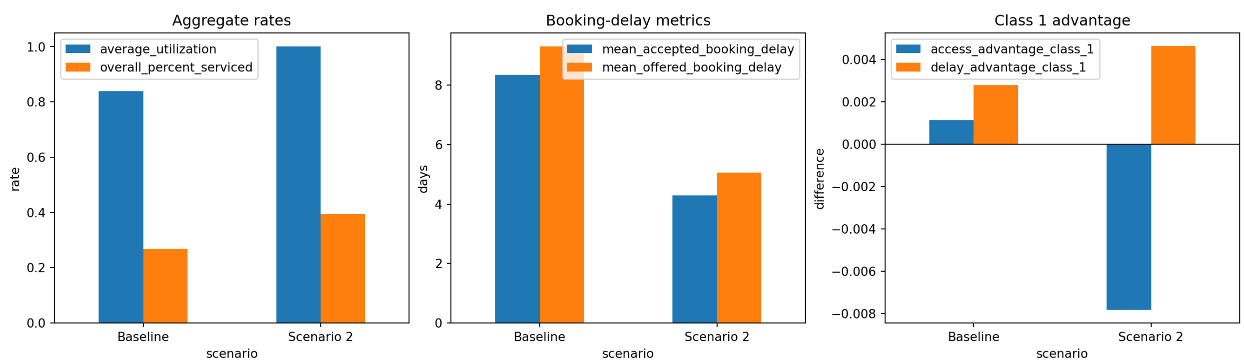


Figure 1: Baseline and Scenario 2 comparison. The main analysis below focuses on the baseline FCFS model and changes one behavior family at a time.

| Scenario | Utilization | Overall served | Accepted wait | Offered wait | Class gap | Delay gap |
|---|---|---|---|---|---|---|
| Baseline | 0.839 | 0.269 | 8.35 | 9.30 | 0.001 | 0.003 |
| Scenario 2 | 1.000 | 0.395 | 4.29 | 5.06 | -0.008 | 0.005 |

Table 2: Scenario values are rates or averages. Class gap is Class 1 served rate minus Class 2 served rate.

In the baseline, the two classes have the same arrival rate and behavior rules. Because of that, any large class difference in the sensitivity plots is caused by the parameter difference introduced in that plot.

# 4 How to Read the Behavior Panels

Most behavior sections use the same six-panel heatmap layout:

- top row: overall served rate, Class 1 served rate, Class 2 served rate;
- bottom row: overall offered wait, Class 1 offered wait, Class 2 offered wait.

The x-axis is the Class 1 value of the behavior parameter. The y-axis is the Class 2 value. The diagonal is where both classes have the same value. Off the diagonal, one class has different behavior from the other.

This layout is meant to answer the practical question directly: does the parameter improve the whole system, or does it mostly move access from one class to the other?

# 5 Balking

Balking means a patient receives an offer but rejects it because the appointment is too far away. In this model, balking has two parts:

- the **threshold**: the delay where high balking begins;
- the **step**: the high balking probability after that threshold.

A higher step means patients reject long-delay offers more often. A lower threshold means patients begin rejecting at shorter waits.

## 5.1 Balking Step

**What changes.** Class 1 and Class 2 balking step values vary from 0 to 1.

**What stays fixed.** Balking thresholds stay at the baseline value of 9 days. All no-show, cancellation, arrival, capacity, and horizon settings stay at baseline.

**What to look at.** The served-rate panels show who remains in the system. The offered-wait panels show whether more balking reduces the wait environment.

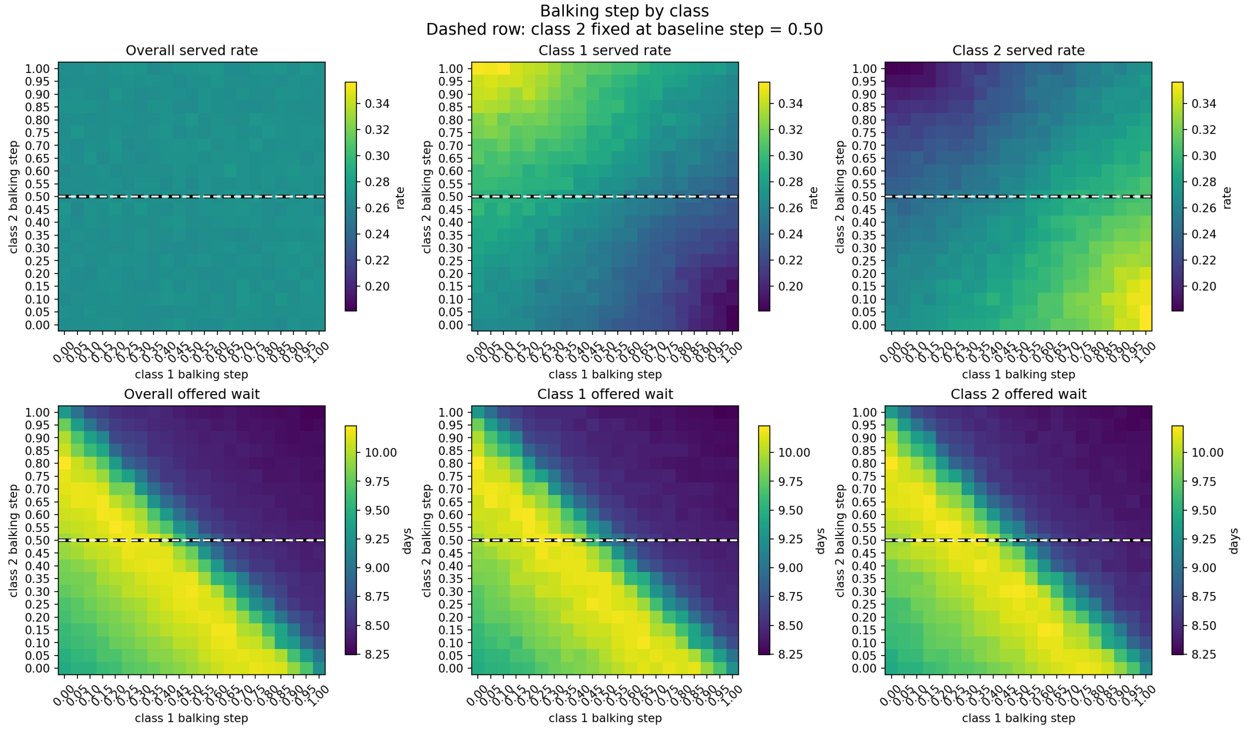


Figure 2: Balking step sweep. Top row shows served rates. Bottom row shows offered waits.

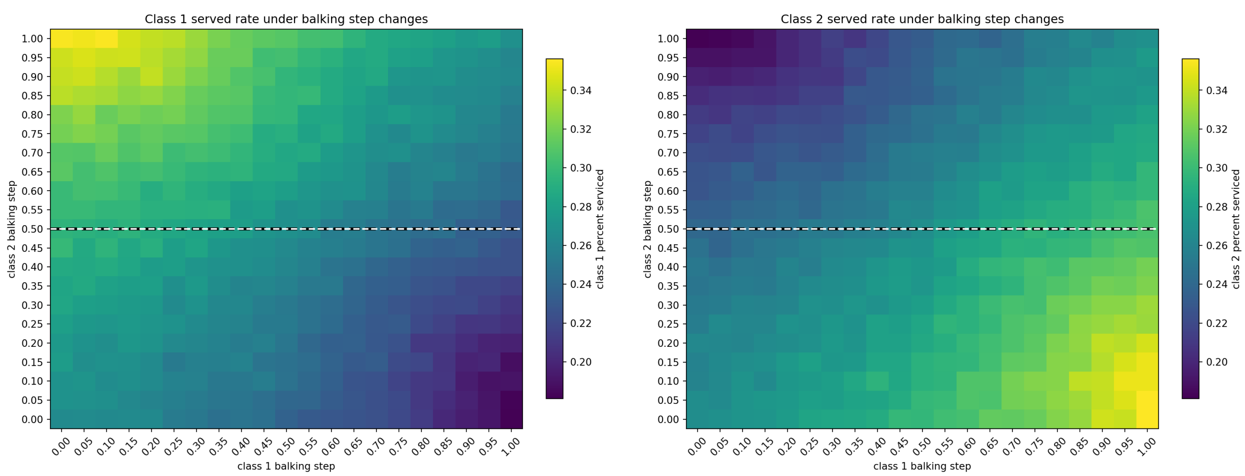


Figure 3: Class-specific served-rate heatmaps for the balking step sweep. Left: Class 1. Right: Class 2.

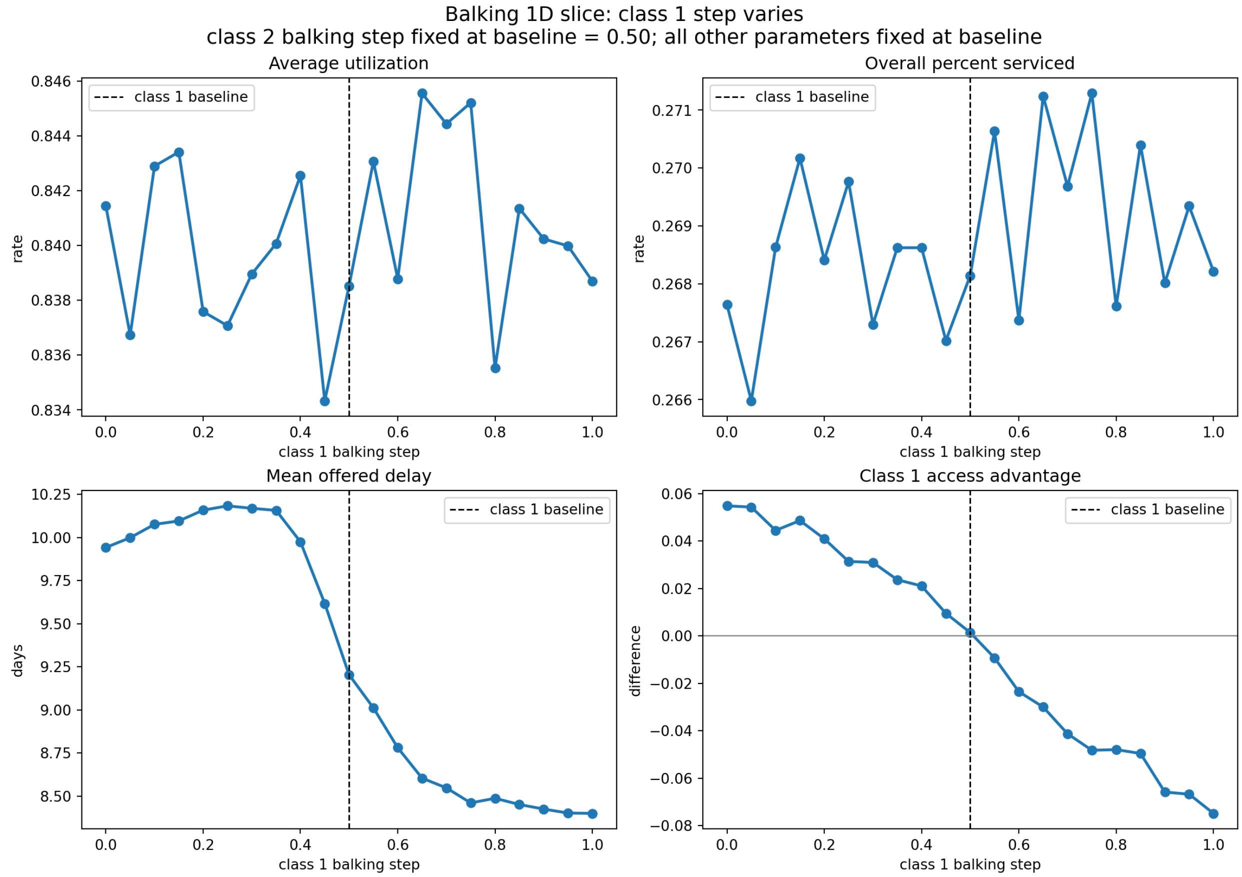


Figure 4: One-dimensional slice: Class 1 balking step changes while Class 2 stays fixed at the baseline step, 0.50.

**Results.** When a class has a higher balking step, that class rejects more long-delay offers. Its served rate falls because more of its arrivals leave before booking. The other class can benefit because released slots become available to it. Offered wait can fall when balking becomes high enough, but lower wait is not always a pure improvement because some patients are leaving the system.

**Main takeaway.** Balking step is mainly a redistribution parameter. The class with the lower balking step is more likely to be served. Overall utilization often stays fairly stable because the clinic still has enough demand to fill slots.

## 5.2 Balking Threshold

**What changes.** Class 1 and Class 2 balking thresholds vary across the booking horizon.

**What stays fixed.** Balking step stays at the baseline value of 0.50. All other behavior and arrival settings stay at baseline.

**What to look at.** A lower threshold means patients start rejecting earlier. A higher threshold means patients tolerate longer waits before high balking begins.

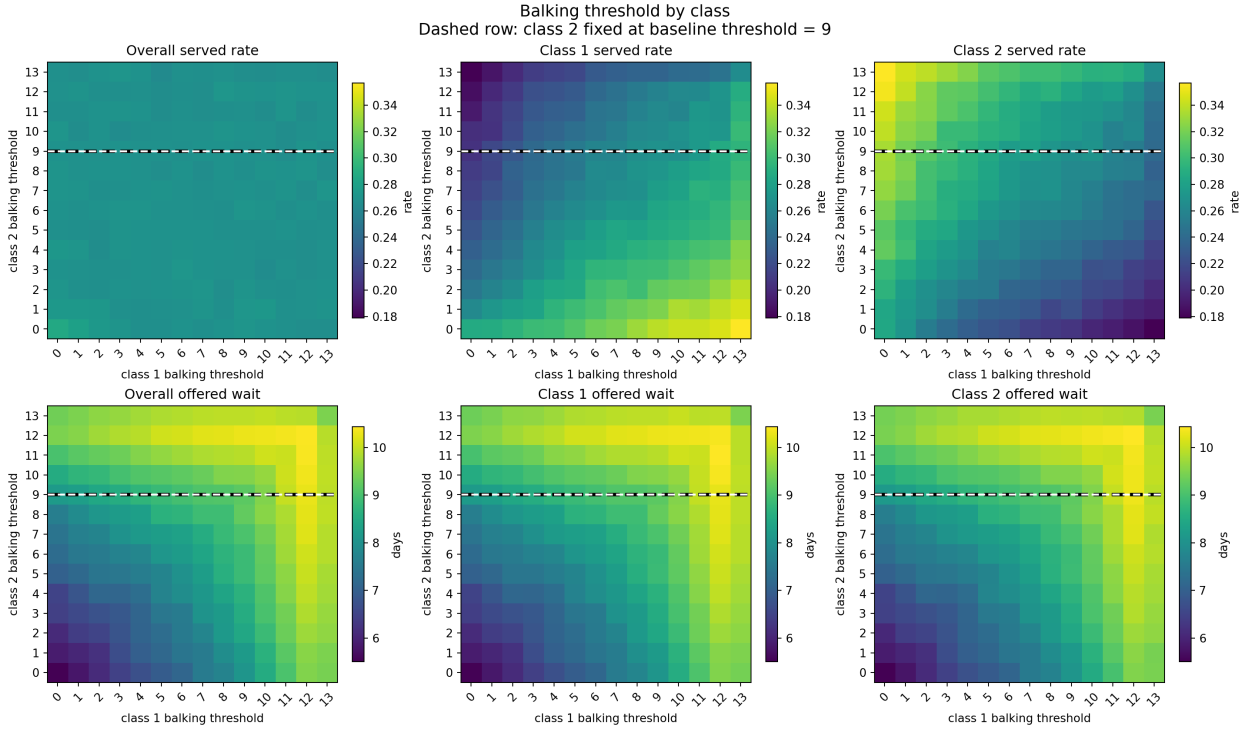


Figure 5: Balking threshold sweep. Top row shows served rates. Bottom row shows offered waits.

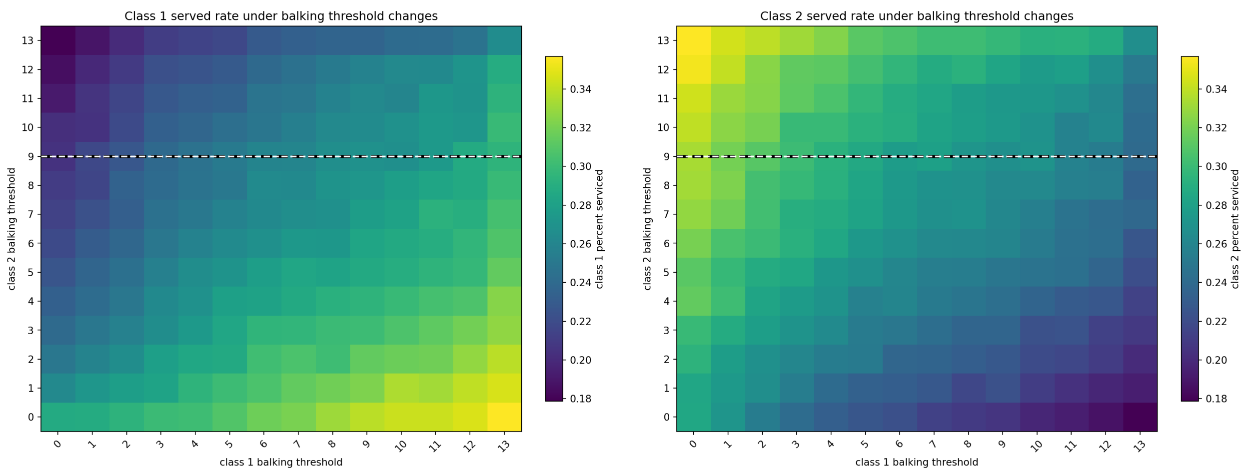


Figure 6: Class-specific served-rate heatmaps for the balking threshold sweep. Left: Class 1. Right: Class 2.

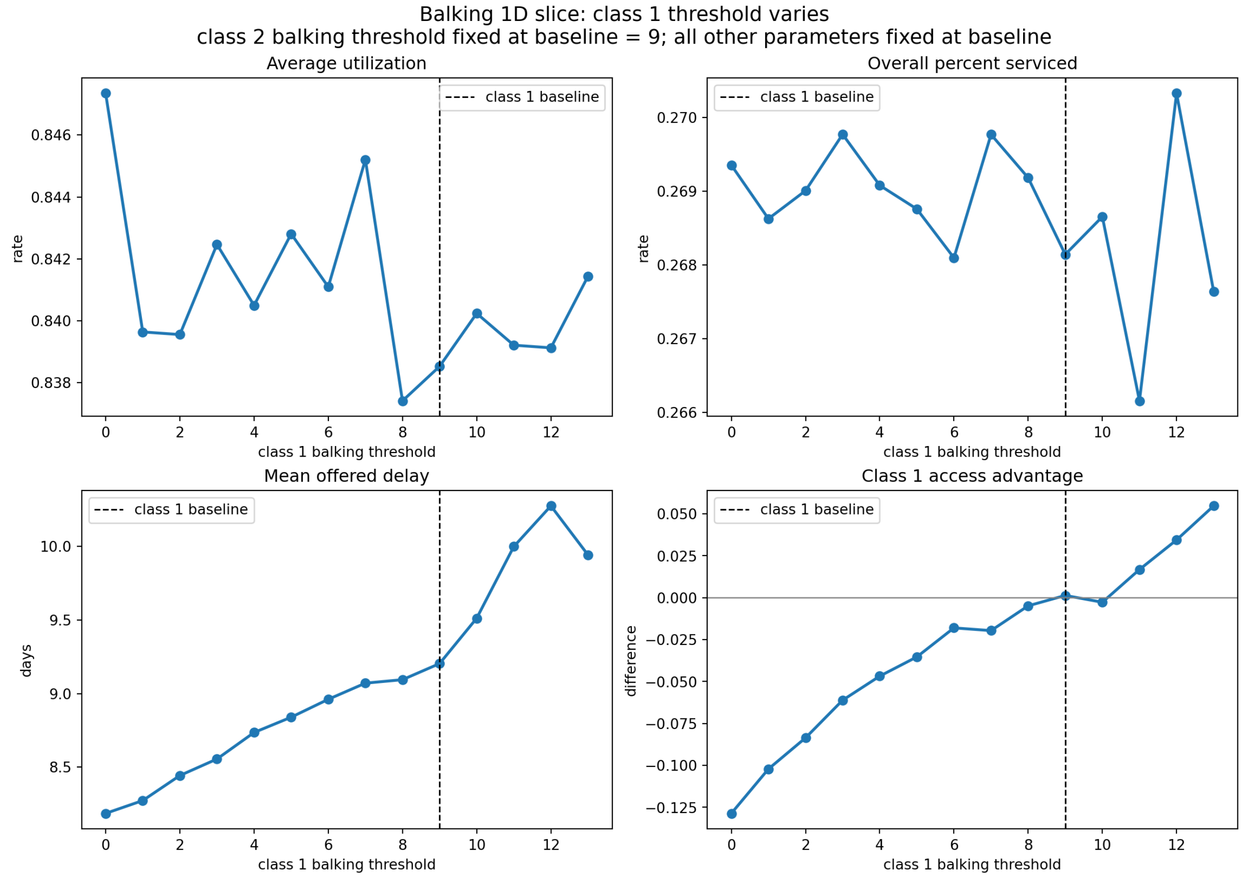


Figure 7: One-dimensional slice: Class 1 balking threshold changes while Class 2 stays fixed at the baseline threshold, 9 days.

**Results.** The class with the higher threshold tends to receive more completed service because its patients accept longer waits. The class with the lower threshold leaves the system earlier. Offered wait can rise when thresholds are high because more patients accept far-future appointments and keep the calendar congested.

**Main takeaway.** Balking threshold is also mainly redistributive. The class that tolerates longer waits is more likely to be served. A lower accepted wait can be misleading if it happens because many long-wait patients balked and disappeared from the accepted-wait calculation.

## 5.3 Balking Threshold and Step Together

**What changes.** Both classes receive the same balking threshold and the same balking step at each grid point.

**What stays fixed.** The two classes are treated identically in this sweep. Arrival rates, no-show, cancellation, capacity, and horizon stay at baseline.

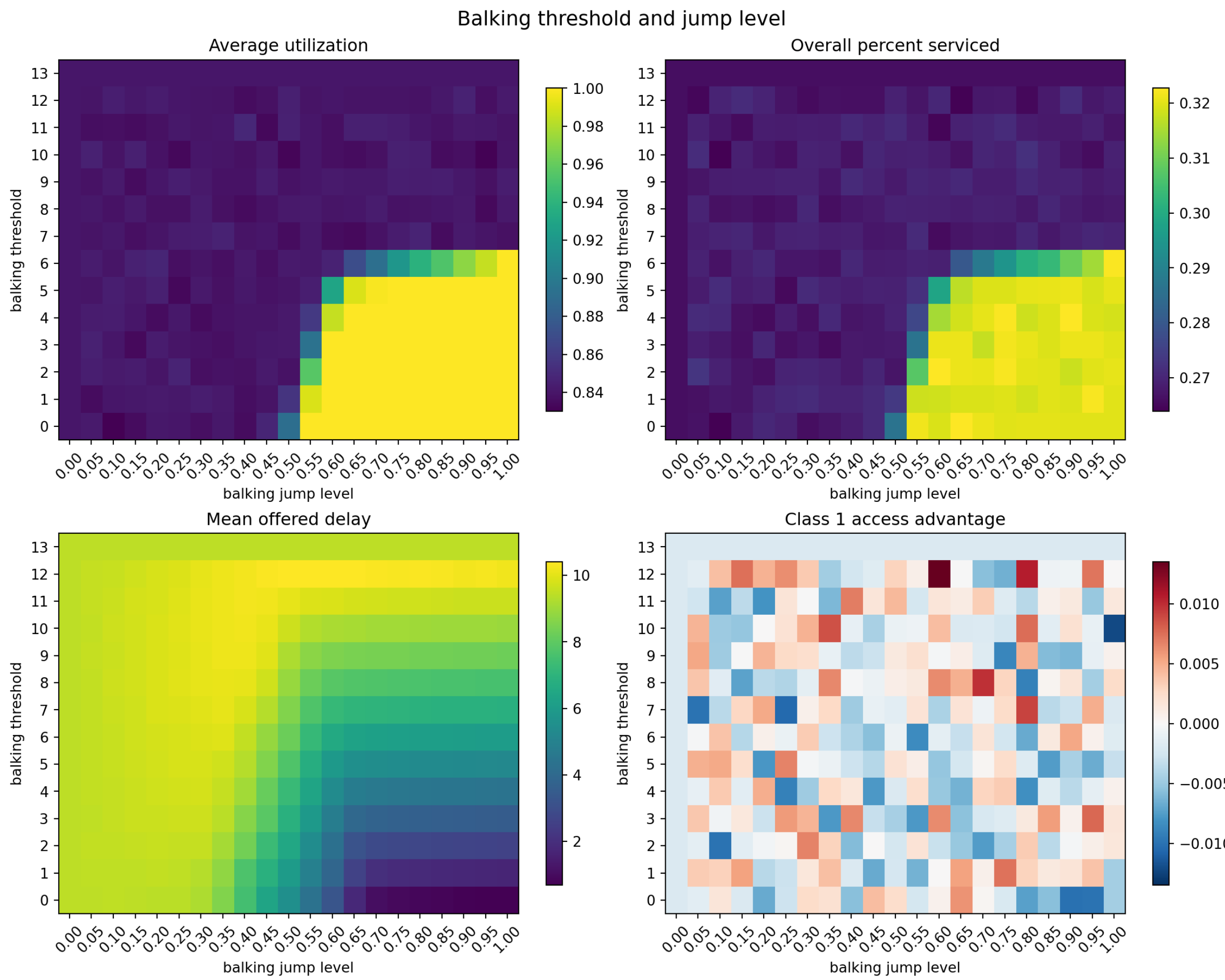


Figure 8: Balking threshold and step when both classes share the same values.

**Results.** Because both classes use the same balking rule, there is almost no structural class advantage. The main effect is system-level: lower thresholds and higher steps remove more long-delay demand, while higher thresholds and lower steps keep more patients in the booking pool.

**Main takeaway.** Absolute balking level affects the access/wait tradeoff. Class advantage appears when the classes differ from each other.

# 6 No-Show

No-show behavior happens after booking. A patient has already accepted an appointment, but then may not attend. This means no-shows affect completed visits and utilization directly, but they do not strongly change the delay that was offered at booking time.

## 6.1 No-Show Step

**What changes.** Class 1 and Class 2 no-show step values vary from 0 to 1.

**What stays fixed.** No-show thresholds stay at the baseline value of 6 days. Balking, cancellation, arrival, capacity, and horizon stay at baseline.

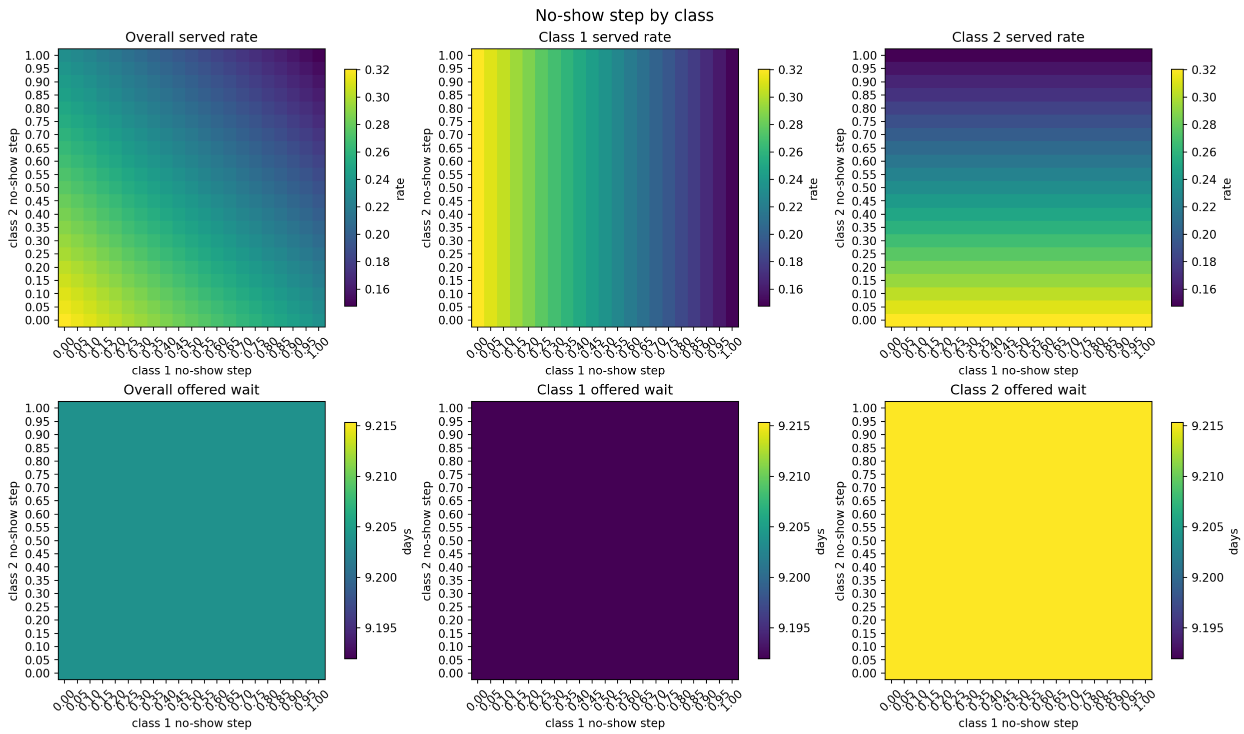


Figure 9: No-show step sweep. Top row shows served rates. Bottom row shows offered waits.

**Results.** A higher no-show step means more booked patients fail to attend after the threshold. That lowers the served rate for the class with higher no-show risk and reduces aggregate utilization because booked slots are lost. Offered wait changes little because the no-show decision happens after the appointment was booked.

**Main takeaway.** No-show step has a strong efficiency effect and a class redistribution effect. It wastes capacity and benefits the class with lower no-show risk.

## 6.2 No-Show Threshold

**What changes.** Class 1 and Class 2 no-show thresholds vary across the booking horizon.

**What stays fixed.** No-show step stays fixed. Balking, cancellation, arrival, capacity, and horizon stay at baseline.

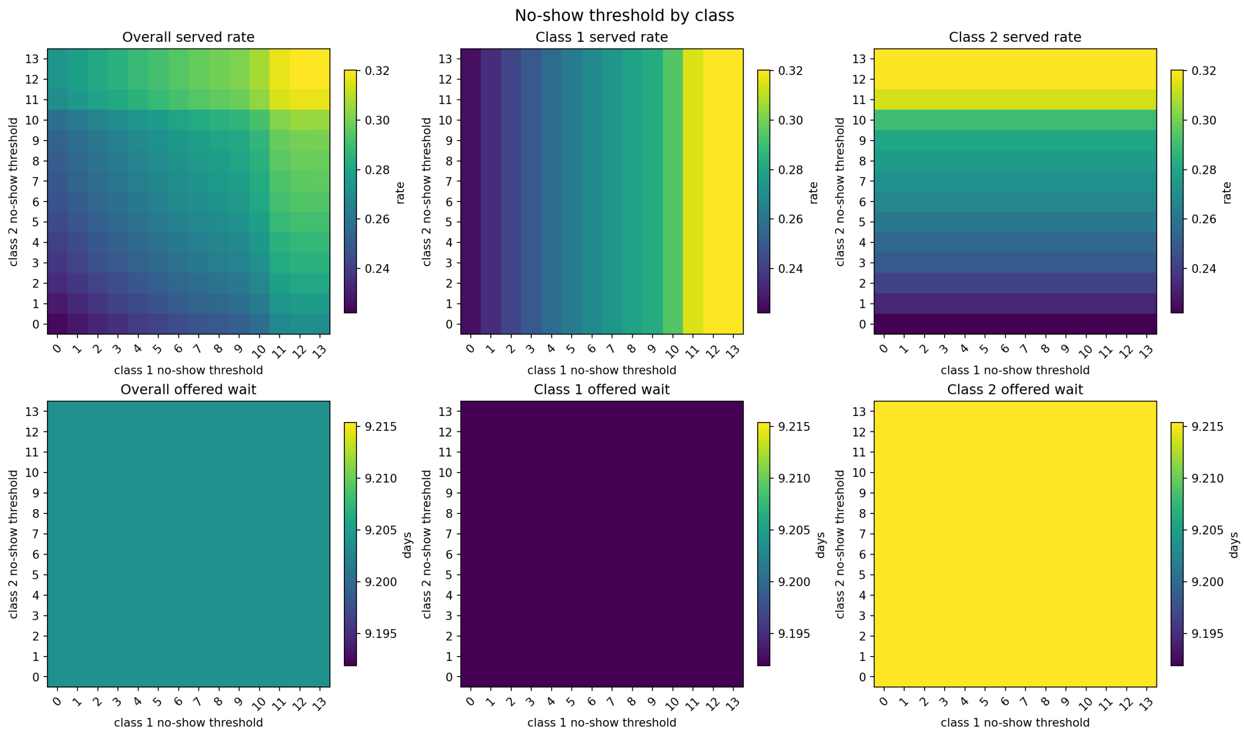


Figure 10: No-show threshold sweep. Top row shows served rates. Bottom row shows offered waits.

**Results.** A higher no-show threshold means no-show risk begins only at longer waits. That improves the served rate for that class and can improve overall utilization. Offered wait remains much less sensitive than served rate because no-show behavior occurs after booking.

**Main takeaway.** The class with the higher no-show threshold is more likely to complete service. No-show thresholds matter for completed visits, not for offered wait.

## 6.3 No-Show Threshold and Step Together

**What changes.** Both classes receive the same no-show threshold and the same no-show step at each grid point.

**What stays fixed.** The two classes are treated identically in this sweep. Balking, cancellation, arrival, capacity, and horizon stay at baseline.

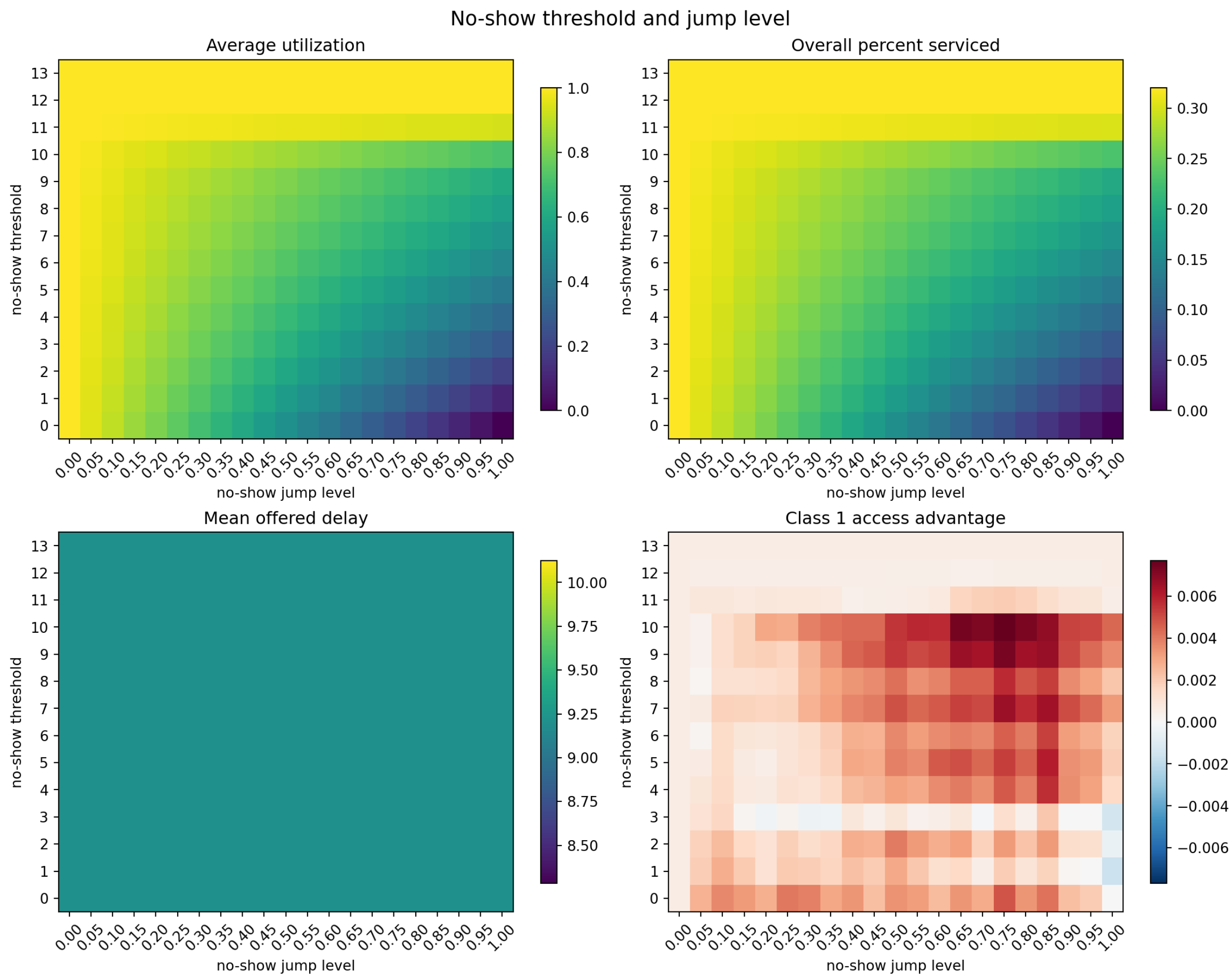


Figure 11: No-show threshold and step when both classes share the same values.

**Results.** High no-show step and low no-show threshold create many missed appointments. This lowers utilization and the overall served rate. Since both classes have the same no-show rule, there is little class advantage.

**Main takeaway.** Absolute no-show behavior is one of the clearest efficiency levers. Class advantage appears when one class has lower no-show risk than the other.

# 7 Cancellation

Cancellation is different from balking and no-show. It is not threshold-based in the current model. It is a daily probability that a future appointment is canceled before the service day.

## 7.1 Cancellation Probability

**What changes.** Class 1 and Class 2 cancellation probabilities vary from 0 to 0.30.

**What stays fixed.** Balking, no-show, arrival, capacity, and horizon stay at baseline.

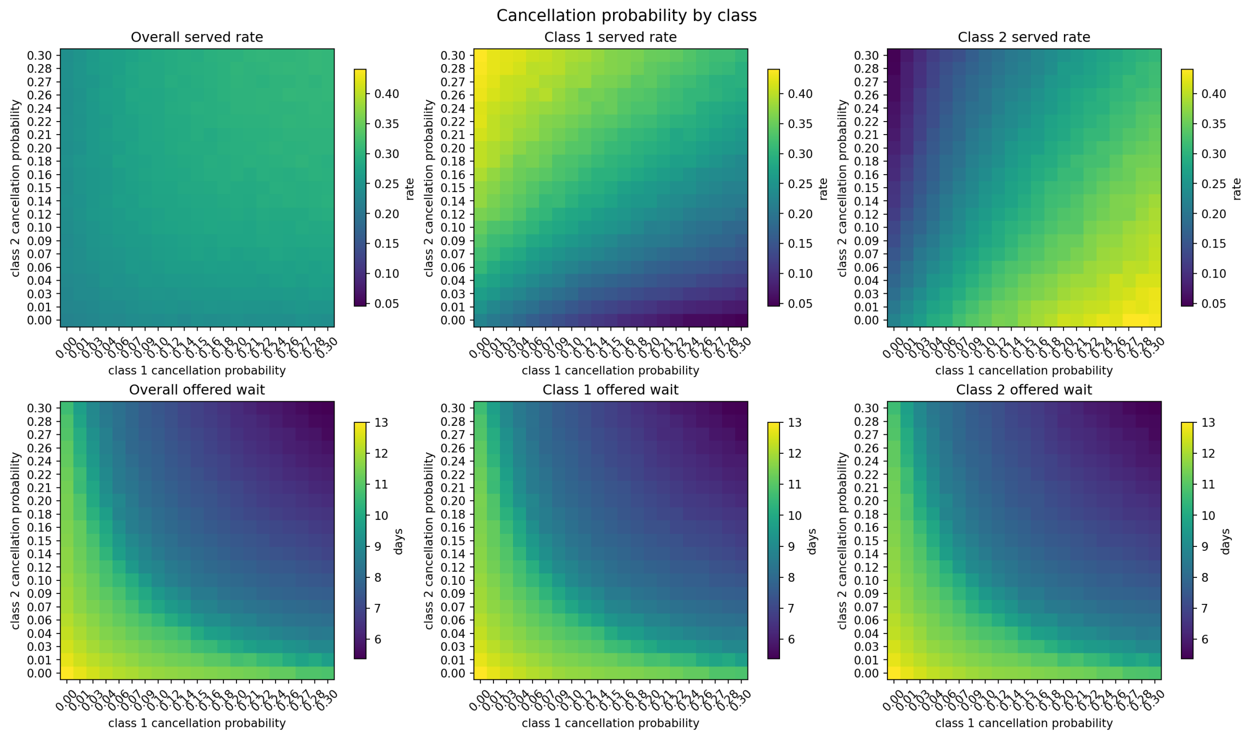


Figure 12: Cancellation probability sweep. Top row shows served rates. Bottom row shows offered waits.

**Results.** A class with higher cancellation probability loses more booked patients before service. Cancellations may free future capacity for someone else, but the canceling patient is still lost. This creates a large class served-rate gap when cancellation differs by class.

**Main takeaway.** Cancellation is a direct completion-risk parameter. It is one of the strongest drivers of class advantage because the class with lower cancellation probability is much more likely to complete service.

# 8 Arrival Rate and Class Mix

Arrival parameters control how much demand enters the system. They do not directly change patient behavior, but they change congestion.

## 8.1 Total Arrival Rate and Class Mix

**What changes.** Total arrivals per day and the Class 1 share of arrivals vary. The report uses actual arrival rates because that is easier to read than a normalized demand scale.

**What stays fixed.** Balking, no-show, cancellation, capacity, and horizon stay at baseline.

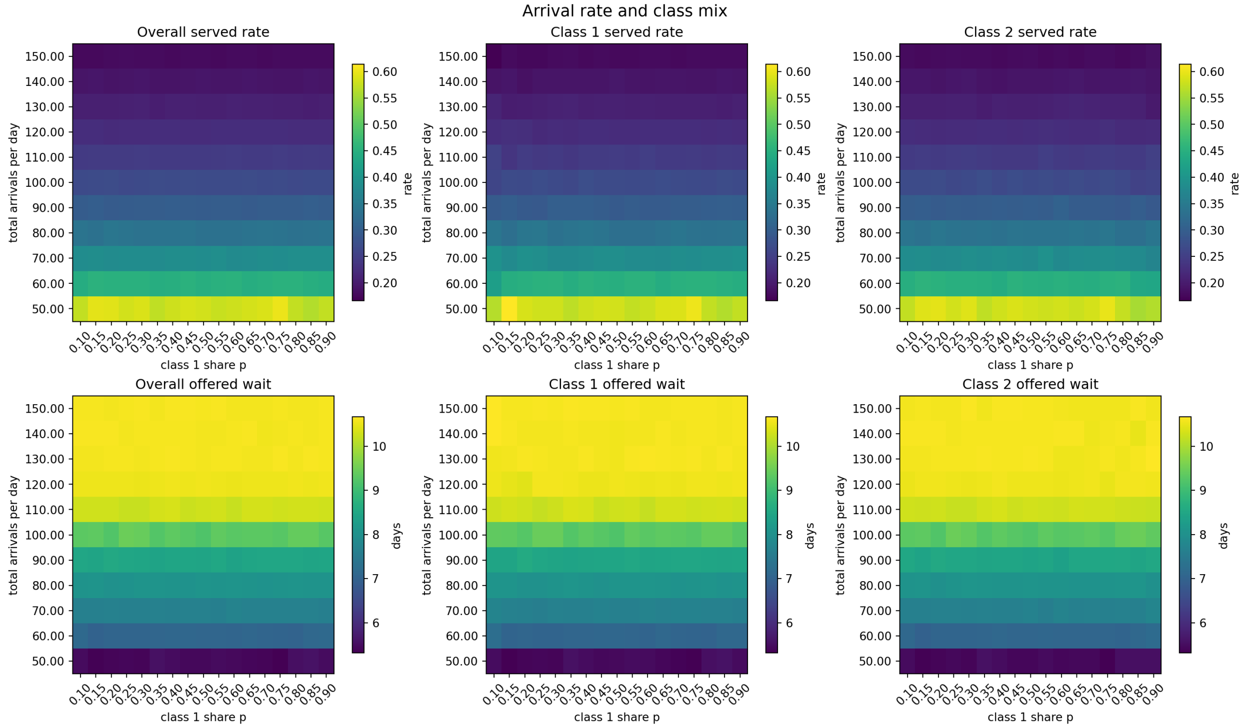


Figure 13: Arrival rate and class mix. Top row shows served rates. Bottom row shows offered waits.

**Results.** Higher total arrival rate lowers the overall served rate and raises offered wait. Utilization can remain high even when access is poor because the clinic is busy but cannot serve a large share of arrivals. Changing the class mix has a weaker class-benefit effect when the two classes have the same behavior rules.

**Main takeaway.** Demand pressure is the main aggregate access driver. Class mix alone mostly changes volume unless the classes also differ in behavior.

## 8.2 Changing One Class Arrival Rate

**What changes.** One class arrival rate changes while the other class stays fixed.

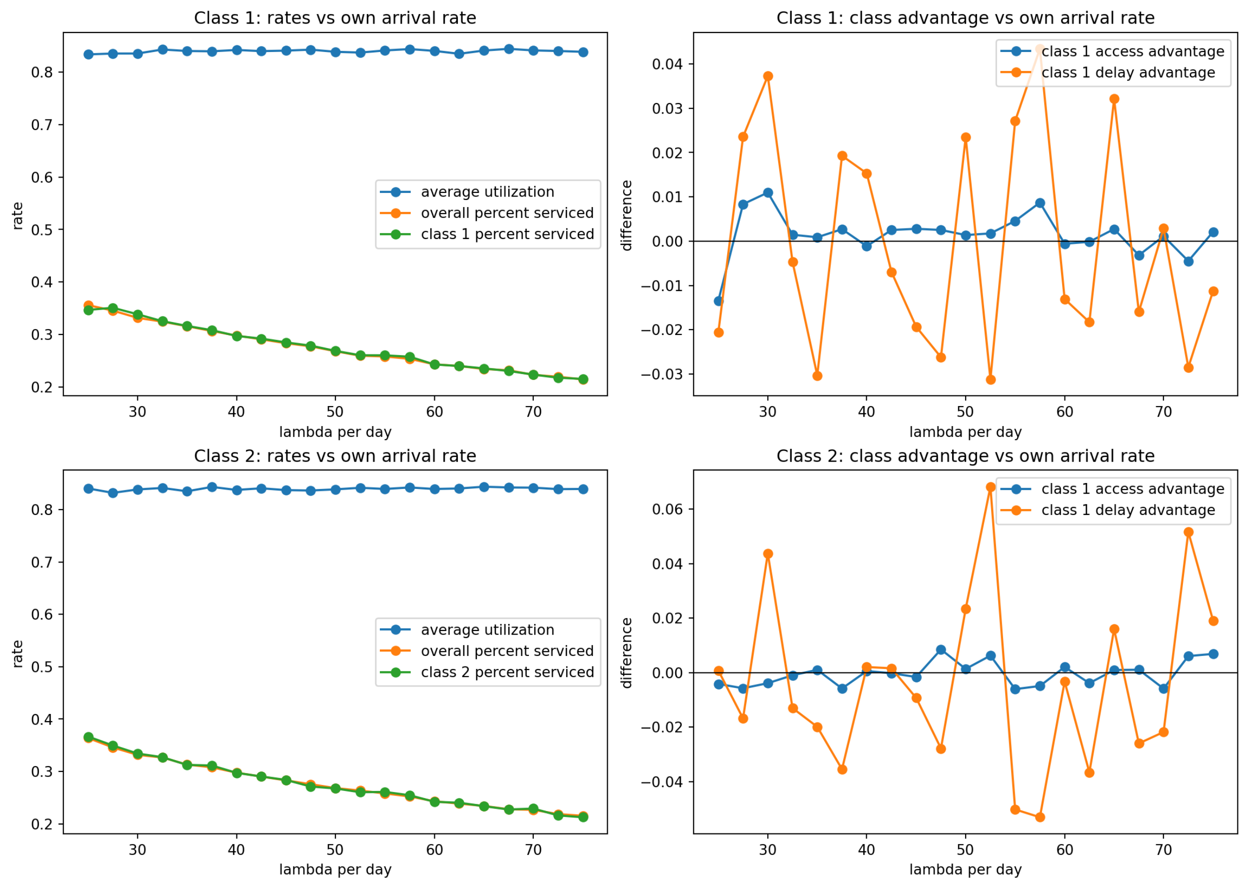


Figure 14: Changing one class arrival rate at a time.

**Results.** Increasing either class arrival rate increases congestion. Under FCFS and equal behavior rules, this does not create a strong persistent class advantage by itself.

**Main takeaway.** Own-class arrival volume matters for congestion, but class labels do not matter much unless class behavior differs.

# 9 FCFS Stress Test

The previous sections change behavior parameters. This section keeps behavior fixed and varies total demand to show how FCFS behaves under light load, baseline load, and overload.

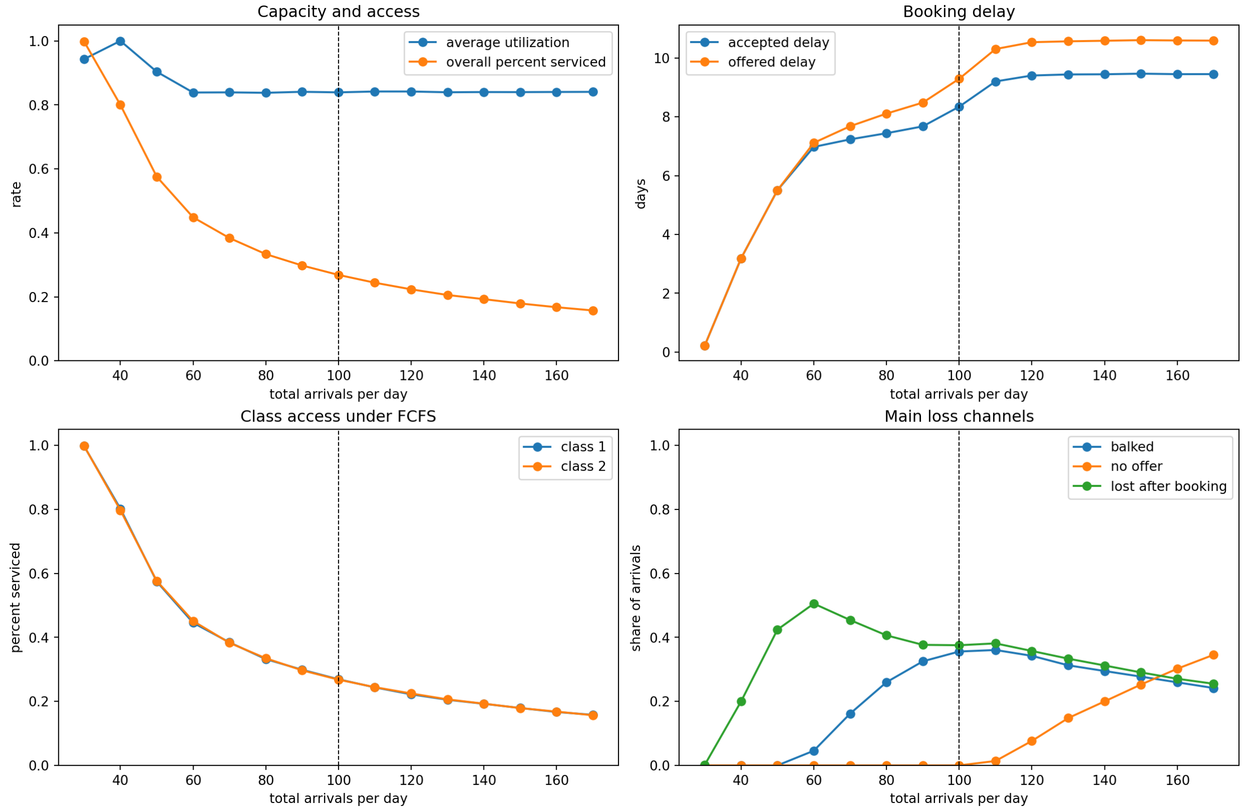


Figure 15: FCFS stress curves as total arrivals per day change. The vertical dashed line marks the baseline total arrival rate.

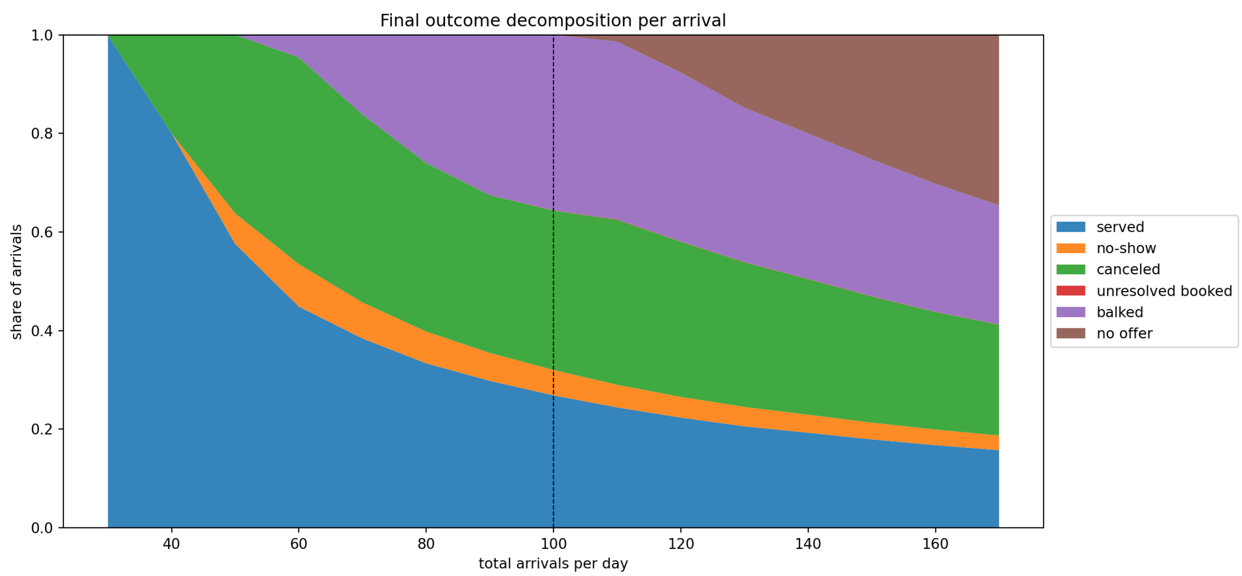


Figure 16: Arrival outcome decomposition under FCFS. Each band is a share of measured arrivals.

| Total arrivals/day | Utilization | Overall served | Offered wait | Served | Balked | No offer | Lost after booking | Class gap |
|---|---|---|---|---|---|---|---|---|
| 30 | 0.943 | 0.998 | 0.22 | 0.998 | 0.000 | 0.000 | 0.002 | -0.000 |
| 100 | 0.839 | 0.269 | 9.30 | 0.269 | 0.356 | 0.000 | 0.376 | 0.001 |
| 170 | 0.841 | 0.157 | 10.59 | 0.157 | 0.242 | 0.346 | 0.255 | 0.001 |

Table 3: Selected stress-test points. Outcome columns are shares of arrivals. The baseline demand level is 100 arrivals per day.

**Results.** Under light load, almost every arrival is served. Near baseline, the clinic is busy, but only a minority of arrivals complete service. Under overload, utilization still looks high, but the served rate falls further and no-offers become a major loss channel.

**Main takeaway.** Utilization alone is not enough. A clinic can be busy and still have poor access. Served rate and outcome decomposition are needed to understand FCFS performance.

## 10 Regression Screening

The heatmaps change one or two parameters at a time. The regression screen asks a broader question: across many random parameter settings, which inputs best explain the outputs?

**What changes.** 240 random FCFS parameter settings were sampled. Each setting was run with 2 random seeds and averaged.

**How to read it.** The regression coefficients are standardized. A larger absolute coefficient means that input has a stronger average association with the target metric over the sampled range.

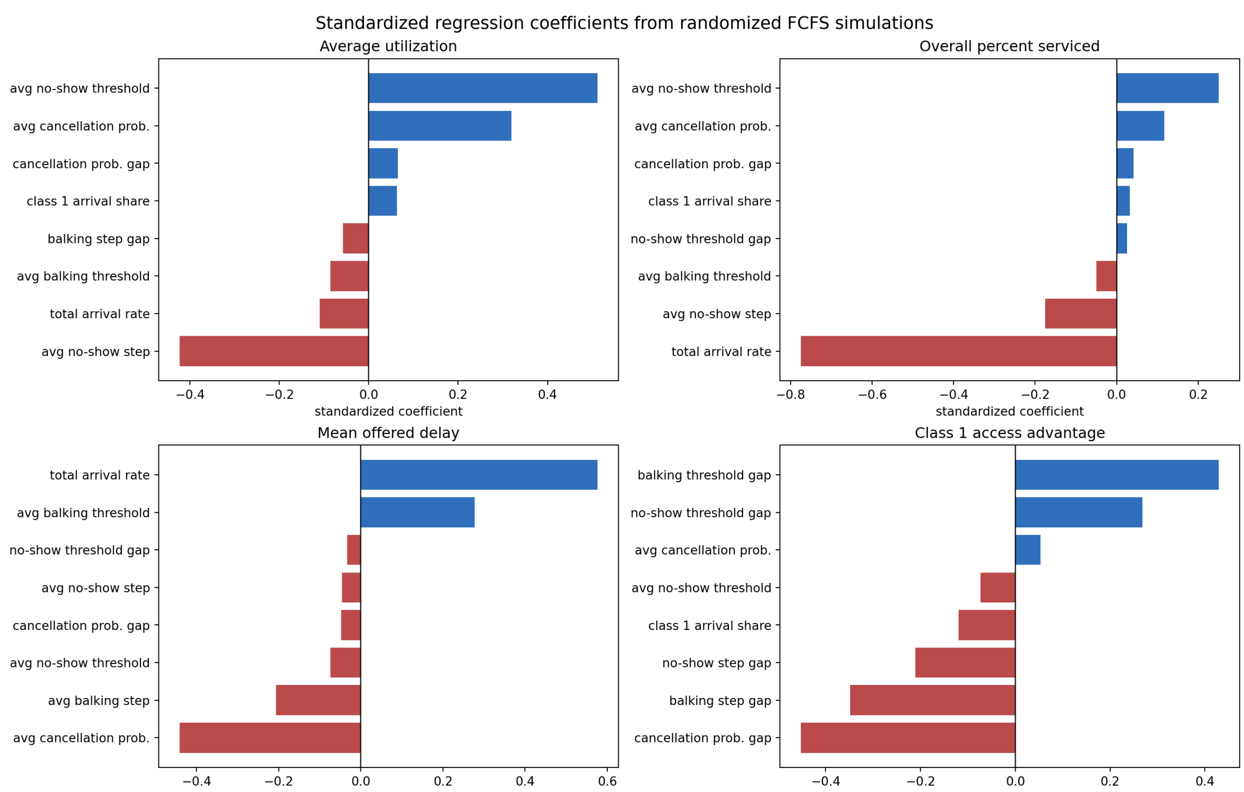


Figure 17: Largest standardized regression coefficients from randomized FCFS simulations. Positive coefficients increase the target metric; negative coefficients decrease it.

| Target | Most important feature | Standardized coefficient |
|---|---|---|
| Utilization | Average no-show threshold | 0.512 |
| Utilization | Average no-show step | -0.423 |
| Utilization | Average cancellation probability | 0.320 |
| Overall served rate | Total arrival rate | -0.774 |
| Overall served rate | Average no-show threshold | 0.251 |
| Overall served rate | Average no-show step | -0.175 |
| Offered wait | Total arrival rate | 0.576 |
| Offered wait | Average cancellation probability | -0.441 |
| Offered wait | Average balking threshold | 0.278 |
| Class gap | Cancellation probability gap | -0.452 |
| Class gap | Balking threshold gap | 0.429 |
| Class gap | Balking step gap | -0.349 |

Table 4: Top regression drivers. Gap means Class 1 value minus Class 2 value.

**Results.** Aggregate performance is mostly explained by total arrival rate and average no-show behavior. Class gap is mostly explained by differences between the classes: cancellation gap, balking threshold gap, and balking step gap.

**Main takeaway.** The regression confirms the heatmaps. Absolute demand and no-show behavior explain overall performance. Differences between class behavior explain who benefits.

## 11 Final Takeaways

1. **Served rate is the main access metric.** Utilization can stay high even when many patients are not served.
2. **Balking mainly redistributes access.** The class that balks less, or tolerates longer waits, is more likely to be served.
3. **No-show behavior is a strong efficiency driver.** No-shows waste booked slots and lower completed visits.
4. **Cancellation differences create large class gaps.** A class with lower cancellation probability is much more likely to complete service.
5. **Arrival rate controls congestion.** Higher demand lowers the overall served rate and increases offered wait.
6. **Class mix alone is weak unless behavior differs.** FCFS does not strongly privilege a class label by itself.
7. **For class benefit, gaps matter more than absolute levels.** If both classes share the same behavior rule, class advantage stays near zero. If one class has a more favorable rule, that class benefits.